

Asteroid Impact Risk Predictor

Business Plan

A Machine-Learning-Powered Planetary Defense Screening Service, built on the NASA JPL Small-Body Database.

Prepared as part of a graduation project in Data Science / AI

1. Executive Summary

Asteroid Impact Risk Predictor is a machine-learning system that screens Near-Earth Objects (NEOs) and flags Potentially Hazardous Asteroids (PHAs) using their orbital and physical characteristics. Built on NASA's own certified Small-Body Database (over 950,000 catalogued objects), the system automates an initial risk-screening step that currently relies on manual review by domain experts, reducing the number of objects that require detailed human assessment.

The product is positioned not as a replacement for official planetary-defense infrastructure (such as NASA's Sentry system), but as a complementary, lightweight screening and educational layer that can serve research institutions, aerospace companies, insurers of space assets, science-education platforms, and government agencies in regions without dedicated planetary-defense programs.

2. Problem Statement

Hundreds of thousands of asteroids are catalogued, with new discoveries added continuously by ground and space-based surveys. Manually assessing every newly discovered or newly characterized object for hazard potential is time- and expertise-intensive. Existing official tools (e.g. NASA/JPL Sentry) are authoritative but are not designed as an accessible, embeddable screening layer for third parties such as universities, smaller space agencies, satellite operators, or insurers who need a fast, automated first-pass risk estimate integrated into their own workflows or products.

3. Proposed Solution / Product

A trained classification model (Random Forest / XGBoost, leakage-audited and threshold-tuned) that accepts an asteroid's orbital and physical parameters and returns a hazard probability score. The product is delivered through three tiers:

- **Public Web Demo:** A free, ad-free Streamlit-based interface for educational use, science outreach, and quick individual lookups.
- **API-as-a-Service:** A subscription REST API for institutions to integrate hazard screening into their own internal tools, dashboards, or research pipelines.
- **Custom / Enterprise Deployment:** A licensed, on-premise or private-cloud version with model retraining on an organization's proprietary tracking data (e.g. a national space agency's own observation network).

4. Market Opportunity

The global space economy is expanding rapidly, and planetary defense has gained institutional attention following recent missions such as NASA's DART. Adjacent markets relevant to this product include:

- Space-situational-awareness (SSA) and space-traffic-management vendors
- Satellite operators and space-asset insurers assessing collision/debris-adjacent risk
- University astronomy/data-science departments needing teaching and research tools

- National space agencies in emerging space nations without in-house Sentry-equivalent systems
- STEM education platforms and science-museum interactive exhibits

5. Business Model & Revenue Streams

Revenue Stream	Customer	Pricing Model
API subscription	Research institutions, aerospace companies	Monthly/annual tiered plans by call volume
Enterprise licensing	National space agencies, satellite operators	Annual license + custom retraining fee
Educational licensing	Universities, science museums	Flat annual fee per institution
Public demo	General public, students	Free (lead-generation / brand awareness)

6. Technology Overview

The system is built on a leakage-free, imbalance-aware machine learning pipeline: data cleaning and missing-value handling (including a dedicated sub-model to estimate missing physical diameters), engineered orbital and physical features, class-imbalance correction, and multiple candidate classifiers evaluated with Precision/Recall/F1/ROC-AUC rather than accuracy alone. The final model is deployed through a lightweight web application and can be exposed as a REST API for institutional integration.

7. Competitive Landscape

Competitor / Alternative	Strength	Gap this product addresses
NASA/JPL Sentry System	Authoritative, official	Not designed as an embeddable third-party API/service
ESA NEO Coordination Centre	Strong European coverage	Similar institutional-only access model
Manual expert review	High accuracy on edge cases	Slow, not scalable to high volumes

8. SWOT Analysis

Strengths	Weaknesses
Built on certified NASA data; leakage-audited methodology	High operational cost (and proprietary telescope network needed)
Opportunities	Threats
Growing space economy and SSA market; emerging space agencies without in-house tools	Official agencies without in-house tools; STEM educational demand; regulatory/liability

9. Product Roadmap

- **Phase 1 (0–3 months):** Public web demo, model documentation, open-source core pipeline for credibility.
- **Phase 2 (3–9 months):** REST API with authentication and usage-based billing; outreach to 2–3 pilot university partners.
- **Phase 3 (9–18 months):** Enterprise/agency licensing track; retraining pipeline for organization-specific tracking data.
- **Phase 4 (18+ months):** Expansion into adjacent screening products (e.g. satellite conjunction/collision risk).

10. Risks & Honest Limitations

This plan should be read alongside the model's real, documented limitations: the hazardous class is extremely rare (~0.22% of the dataset), and even after threshold tuning the model's Recall on true hazards is approximately 72%, meaning a meaningful share of true positives can be missed. The product must be marketed strictly as a *screening and decision-support* tool, never as a certified hazard-determination authority, to manage liability and to stay factually honest about what the model can and cannot guarantee.

11. Conclusion

By packaging a scientifically grounded, transparently evaluated machine learning pipeline into an accessible product, this project has a credible path from an academic exercise to a real screening tool for research institutions, emerging space programs, and education — while remaining honest about its role as a complement to, not a replacement for, official planetary-defense systems.